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SUPERSEDING
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MILITARY SPECIFICATION

CLEANER, STEAM, PRESSURE JET, WHEEL-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two sizes of electric-motor-driven wheel-mounted, oil-fired, pressure jet, steam cleaners. The cleaners combine cleaning compound, heat, and water under pressure to produce jet-cleaning solution that is ejected at a minimum of 100 gallons per hour (gph) and 180 gph.

1.2 Classification. Steam cleaners shall be of the following sizes, as specified (see 6.2):

Size	Minimum rated capacity (gph)
1	100
2	180

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-C-437 - Cleaning Compound, High Pressure (Steam) Cleaner.
WW-C-624 - Coupling Assembly, Hose (Garden, Water, and Water Suction).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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ZZ-H-561 - Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
ZZ-H-601 - Hose and Hose Assemblies, Rubber (Yarn or Fabric Reinforced) Water Service.

MILITARY

MIL-S-12651 - Sprayers, Packaging of.
MIL-C-22542 - Cleaning Solution, High Pressure, Liquid.
MIL-H-28596 - Hose Assembly, Rubber, Wire Reinforced 200 psig, Saturated Steam Service.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

390.03 - AGMA Gear Handbook - Volume 1 - Gear Classification, Materials and Measuring Methods for Assembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers Association, 1901 North Ft. Myer Drive, Suite 1000, Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B2.4 - Hose Coupling Screw Threads.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

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AMERICAN PETROLEUM INSTITUTE (API)

Applicable API Gravity/Temperature Standards.

(Application for copies should be addressed to the American Petroleum Institute, 2101 "L" Street, N.W., Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D2156 - Smoke Density in Flu Gasses from Burning Distillate Fuels, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Applicable Codes, Standards, Specifications, and Books on Welding Published or Recommended by ASME.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

JOINT INDUSTRIAL COUNCIL (JIC)

EMP-1/EGP - Electrical Standards.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

MG 1 - Standards for Motors and Generators.
WD 1 - General Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 "L" Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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UNDERWRITERS LABORATORIES INC. (UL)

UL 62 - Flexible Cord and Fixture Wire.
UL 372 - Primary Safety Controls for Gas- and Oil-Fired Appliances.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The steam cleaner (referred to hereinafter as "cleaner") shall consist of a frame, water, soap, fuel, and electrical systems and all controls, gages, and a vapor-cleaning gun. Size 1 and 2 cleaners shall be equipped with wheeled-undercarriage for portability.

3.1.1 Standard commercial product. The cleaner shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the cleaner being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish one cleaner for first article inspection and approval (see 4.2.1 and 6.3).

* 3.3 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

* 3.3.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. When such contact cannot be avoided, joints between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets, or other suitable means. Bolts, nuts, pins, screws, and other fastenings shall be of the same material as the metals joined or shall be cathodic to the metals joined.

* 3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.5 Design. The unit shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment. The cleaner shall be designed for a portable cleaning operation, and shall remove dirt, grease, and oil from vehicles, accessories, and ground equipment by means of a vapor spray composed of a mixture of steam and hot water, with or without cleaning compound.

3.6 Construction. The cleaner shall be complete so that when installed, it can be used for the operations specified herein. The cleaner shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and operation. Construction of its instruments and gages, as such, shall provide for accuracy of measurement and reading. Working parts, as applicable, shall be protected from dirt, snow, and ice.

3.6.1 Safety. All parts which are subject to high operating temperatures, or which are energized electrically, shall be insulated, fully enclosed, or guarded when such parts are exposed to contact with personnel or otherwise would create a hazard. All moving parts hazardous to personnel shall be enclosed or guarded. Exhaust or discharges from the cleaner shall be directed so that they do not endanger personnel. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Protective devices shall not impair the operating functions. Warnings shall be mounted on or near components containing hidden hazards. Steam cleaner noise levels shall not exceed 85 decibels when measured on the A scale of a standard sound level meter at slow response at a height of 5 feet and no greater than 6 feet away from the cleaner. If the combined noise level during any mode of operation measured at the operators position, no greater than 10 feet from the cleaner, exceeds the 8-hour exposure level of OSHA Health and Safety Standard (Title 29 CFR 1910.95), a warning plate shall be attached to the cleaner in a location conspicuous from the control panel stating that hearing protection is required when operating the cleaner.

3.6.1.1 Safety devices. Each cleaner shall be furnished with motor overload protection, a pressure-relief valve, a fuel cutoff, primary safety control, manual valves, and a low water cutoff, and any other required safety devices. Protection against electric shock shall be provided. Closing of the vapor gun valve shall shut off the supply of fuel, water, discharge, and dispensing of cleaning detergent.

3.6.2 Flammable liquid containers and solution tanks. The fuel tank shall be so located and ventilated as to preclude any possibility of fires or explosions caused by the contact of fuel with heated surfaces or electrical apparatus. Solution tanks and solution handling equipment shall be so located and baffled as to prevent spillage of solution onto other parts of the cleaner.

3.7 Metal cabinet. When specified (see 6.2), a metal cabinet with a lock shall be furnished. The metal cabinet shall fully enclose the cleaner components, with exception of the operating valve handle, water inlet connections, discharge outlets, gages, and the electric outlet. Working parts shall be accessible through convenient doors or movable panels. Gages shall

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be flush-mounted in the control panel. The cabinet shall be designed and constructed to resist weather and to prevent water from collecting within or on the cabinet. All normally operated valves and controls shall be accessible from the operator's side.

3.7.1 Stowage. Stowage devices shall be provided for stowing technical publications, tools, and nozzle tips. The stowage devices shall be arranged so that stowed items cannot be lost or dislodged during transit.

3.7.2 Racks. Racks shall be attached to the cleaner for stowing the cleaning gun assembly and all hoses when not in use. A rack shall be provided to house a liquid cleaning solution container of 5-gallon size, when the cleaner employs a liquid chemical cleaning solution system.

3.7.3 Metal thickness. Frame and superstructure shall be fabricated from not less than No. 16 gage sheet metal. A minimum thickness of No. 21 gage sheet steel shall be used for the side paneling. The fuel tank and solution tank shall be constructed of not less than No. 18 gage sheet metal or polyethylene type material. The fuel tank shall have a means for draining sediment and other contaminants.

3.7.4 Gears. All spur gears and pinions used in the machine and its components shall be machined. Where helical, herringbone, bevel, or worm gearing is used, the dimensions and tolerances shall conform to AGMA 390.03.

3.8 Maintainability. All major assemblies shall be accessible for maintenance, repair, and replacement without the removal of assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage. All fasteners shall be of corrosion-resistant material. All screw threads shall be in accordance with FED-STD-H28.

3.9 Mobility. Size 1 cleaners shall have an undercarriage with a minimum of two 8-inch by 2-inch size solid or pneumatic-tired wheels on a common axle, and a supporting leg(s) on the other side of the unit with manufacturer's standard handle bar(s). Size 2 cleaners shall have an undercarriage on a minimum of four 10-inch by 2.75-inch size solid or pneumatic-tired wheels. The rear two wheels of size 2 unit may be mounted on a common axle. The front two wheels of size 2 unit shall be mounted on a swivel with a tee-shaped towing bar that swings easily when towed by hand (see 4.4.3.5).

3.10 Performance.

3.10.1 Cleaner capacity. The cleaner shall be fitted with a vapor-cleaning gun, a 25-foot discharge hose, and provided with cleaning solution conforming to P-C-437 or MIL-C-22542. The cleaner shall have the minimum capacity and pressure range shown in table I as required for operation.

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TABLE I. Cleaner capacity.

Size	Minimum capacity (gph)	Operation pressure range psi
1	100	60 to 100
2	180	75 to 125

3.10.2 Heater efficiency. When operating under continuous load, the vapor generator shall have an overall thermal efficiency of not less than 70 percent (see 4.4.3.2) and emissions shall not exceed No. 2 on the Bacharach scale, (see ASTM D2156).

3.10.3 Burner generating capacity. The burner shall heat within 5 minutes, cold water (40 degrees Fahrenheit (°F) to 55°F) to within 10°F of the saturation temperature at the rated operating pressure and capacity.

3.10.4 Operating pressure, volume, and impact. Unless otherwise specified (see 6.2), the cleaner shall be capable of operation at the pressure range as specified in table I. At constant rated pressure, the cleaner shall deliver the rated volume and develop the minimum impact forces indicated in figure 1 (see 4.4.3.3). The rated pressure of the cleaner shall be marked on the unit. The cleaner solution system shall be capable of withstanding a static pressure of two times the maximum working pressure (see 4.4.3.6).

3.10.5 Storage and operating temperatures. The cleaner, when drained, shall withstand storage without damage in a temperature range of -40°F to +125°F, and shall operate effectively in a temperature range of +40°F to +125°F ambient temperature. The cleaner shall operate effectively at altitudes up to 2,000 feet without requiring readjustment and with minor air adjustment up to 6,000 feet.

3.10.6 Water supply. Each cleaner shall be equipped to secure water from any of the following sources:

a. Tap-pressure system operating from 25 pound-force per square inch (psi) to 125 psi.
b. For size 2 only, when specified (see 6.2), the water system shall be provided with valves and hose (see 3.25) to allow pumping water from tank, well, steam, or other source. The pump suction shall be capable of lifting water a minimum of 6 feet during operation at full cleaner capacity.

3.11 Electrical equipment. Unless otherwise specified (see 6.2), all electrical components including the motors, starters, relays, switches, and wiring shall conform to NFPA No. 70 and shall conform to, and be located in accordance with, JIC EMP-1/EGP. Size 1 cleaners shall operate on 120 volt (V), 60 Hertz (Hz), single phase, electric power. When specified (see 6.2), size 2 cleaners shall operate on 240V, 60 Hz, single phase, electric power. The conductors and electrical components shall withstand an electrical potential of no less than 1,000V root mean square (RMS) at 60 Hz to ground

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without breakdown when tested as specified in 4.4.3.7. All wires, conduit, and electrical components shall withstand the temperatures of normal operation. All wiring and conduits shall be secured to prevent chafing and contact with hot components. There shall be no exposed electrical connections.

3.11.1 Electric motors. Electric motors shall be rated for 120/240V, 60 Hz, single phase operation. Motors shall conform to NEMA MG 1. They shall have sealed ball bearings, sealed windings with class B insulation or better, dripproof construction, and be designed for continuous operation.

3.11.2 Motor switch. A weatherproof, fuse type, manually operated switch shall be provided on the cleaner for electric connection to the cleaner. The switch shall be easily accessible from the operator's side of the cleaner.

3.11.3 Power cord. A UL approved power cable conforming to UL 62 shall be furnished. Power cable shall be three conductor, type SO with stranded copper conductors, size 14 American Wire Gage or larger, and length shall be manufacturer's standard length. Length shall not permit plug end to lay on ground and be contaminated with water. The cord shall be provided with a flexible strain relief at the end attached to the cleaner. The plug end of the cord shall be terminated in either a NEMA WD 1, type 5-15P or NEMA type 6-15P, as applicable, UL approved, 2-pole, 3-wire grounding type plug with dead-front weatherproof construction.

3.12 Heating coil. The heating coil shall be of the continuous-coil type, welded steel construction, and capable of withstanding a hydrostatic test pressure of not less than 1,000 psi. The heater coil tubing shall have a minimum inside diameter (id) of 1/2 inch. The heating coil shall be designed and arranged to permit deliming and easy removal for repairs and replacement, and to prevent formation of excessive carbon deposits on any part of the coil. The steam generator shall contain no drums or extended pressure chambers. Suitable shutoff valves shall be provided at the boiler inlet and outlet for reverse flushing of the boiler, and means shall be provided to keep heated water backflow into supply line in case of water supply failure.

3.12.1 Blowdown. Cleaners shall have the capacity of a forward blowdown cycle. The blowdown shall remove scale and deposits from the heating coils and increase the time between descalings by allowing the heat to remain on for a time with no water supply to the coil. The heat dries and cracks the deposits. A full volume of water is then introduced into the coil flashing to steam which creates sufficient pressure to loosen scale and sludge and flush it out of the coil. Blowdown controls and steam outlets shall be designed and located to preclude operator injury and cleaner damage. Blowdown instructions and safety notes shall be permanently attached to the cleaner, near to and readable, at the blowdown controls.

3.12.2 Insulation. The heating coil shall be jacketed and insulated to prevent hot surfaces from causing injury to personnel.

3.13 Valves. Valves and fittings shall permit complete drainage, blowdown, and reverse flushing. The heating coil shall be equipped with accessible connections and fittings to permit the introduction and circulation of a scale-removing solution of muriatic acid containing 2 percent (by weight) of paraformaldehyde inhibitor. An auxiliary pumping medium is not required.

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3.14 Pressure gages. Dial-type pressure gages of a 2-inch minimum diameter, reading from 1 to 200 psi, shall be attached to the steam outlet side and also to the outlet side of the fuel pump. The gages shall be dampened against vibration or fluctuation. The gages shall be of such quality as to be durable and dependable in service under continual pulsating pressure conditions.

3.15 Burner. The burner shall be of the pressure-atomizing or air-atomizing type, equipped with fuel pump, fan, continuous ignition, and controls. The burner shall be capable of burning any fuel in the classification of light diesel fuel or kerosene (30 to 40 API gravity). A manually operated shut off valve shall be installed in the fuel supply line.

3.15.1 Primary safety control. The oil burner shall be equipped with a primary safety control (flame sensor) that senses the presence or absence of flame, in the event of ignition failure or unintentional flame extinguishment shall cause the burner to shut down. Operation of the burner and the fuel flow shall be terminated and require manual reset for reignition if a flame is not established within 45 seconds, insufficient air is supplied for combustion, or the flame becomes erratic. The flame sensor shall be the heat- or flame-sensing type, and shall not be aided by reflective devices, and shall not be the spark-sensing type. The primary control shall conform to UL 372.

3.15.2 Burner location. The burner shall be so located that openings where fuel or products of combustion might escape are 34 inches above ground level to minimize the danger of vapor explosion. The burner shall be installed in a manner to facilitate cleaning and replacement. The burner shall be designed and located to preclude the possibility of condensate dripping on the burner components.

3.15.3 Ignition for burner system. The ignition system shall provide a continuous spark of sufficient intensity and capacity to insure positive ignition of the fuel used during all conditions of operation. Wiring for the ignition system shall be weatherproof in accordance with NFPA No. 70.

3.16 Fuel limit device. The fuel system shall be equipped with an antidribble device and safety shutoff. When the fuel pressure or flow is insufficient for total atomization, the shutoff shall completely stop the flow of fuel to the burner.

3.17 Automatic safety control. An automatic safety control shall be provided to shut off the flow of fuel in the event of water failure during operation, during any period of excessive coil temperature or pressure, or in the event of flame failure.

3.18 Combustion chamber. When cleaner is equipped with a combustion chamber, the chamber shall withstand without damage, the burning of the fuel/air mixture introduced into the combustion chamber. The combustion chamber shall be purged for 15 seconds before each start. The combustion chamber shall be protected against flame impingement.

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3.19 Fuel tank design. The fuel tank shall be subject to atmospheric pressure only. The tank shall be sealed to prevent possible seepage, splashing, or leakage of other liquids that might contaminate the fuel in the tank. The fuel tank shall have a sufficient capacity for a minimum of 4 hours operation.

3.19.1 Fuel tank accessories. The fuel tank shall be provided with a filling inlet suitably capped and vented. The filler-neck shall be so designed as to preclude any spillage from running into machine components. A strainer shall be in the filler-neck. The tank shall be provided with a drain valve or plug. The fuel line shall have a suitable strainer located betweeen the fuel tank and fuel pump and shall be equipped with a shutoff valve between tank and strainer.

3.19.2 Fuel tank location. The fuel tank shall be so located as to prevent fuel from dripping on the motor, transformer, or electric wiring.

3.20 Fuel pump. The fuel pump shall have a maximum capacity of 4 gph at 130 psi, be packless, and have an adjustable bypass to the fuel tank when a gear driven pump is used. Diaphragm pump application shall not require an adjustable bypass.

3.21 Water pump. The pump shall be a positive displacement type capable of handling either water or cleaning solution at a constant rate of delivery.

3.21.1 Pump construction. The pump shall be of inherently sealed construction to insure constant maximum capacity under continuous operation. The pump shall have all intake and discharge valve parts, such as disks, balls, seats, and springs made of Monel or of stainles steel. The intake and discharge valve parts shall be of such a design that they can be readily replaced. The pump material shall be resistant to the cleaning acid solution and cleaning compound.

3.21.2 Shock-absorbing device. Units provided with a reciprocating or impulse type pump shall be equipped with a shock-absorbing device.

3.22 Cleaning solution systems. When equipped with a liquid type chemical cleaning system the cleaner shall include a rack to secure a 5-gallon container of liquid cleaner conforming to MIL-C-22542. The cleaner shall be designed to inject concentrated liquid cleaning compound into the suction side of the pump or at the coil outlet, at an operator adjustable rate of from 0 to 10 percent by volume. The cleaning compound tank shall have a capacity of no less than 5 gallons and shall be fabricated of stainless steel of the same type as the water supply tank or of high density polyethylene no less than 0.100 thick. The tank shall be vented and equipped with a drain and a gage that can be readily seen by the operator when filling the tank. If the tank is made of polyethylene, it may be transparent or tanslucent in lieu of the gage. The tank shall be located so that the cleaning compound does not drip or splash onto the burner or any electrical components. The cleaning solution tank shall be marked in the same manner as the fuel tank and shall convey the message "Use Liquid Cleaning Compound Conforming to MIL-C-22542". The metering valve to adjust the rate of cleaning compound injected, shall be located on the

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control panel. The metering valve shall indicate the ratio in 0.5 percent increments and shall be marked at intervals of 1 percent. The cleaning compound tank shall have a filler opening no less than 2 inches id and shall have the cap securely chained or cabled to the tank or cleaner.

3.22.1 Powder type solution system. When equipped with a powder type (conforming to P-C-437) chemical system the solution tank shall be located and constructed so that the solution will not drip or splash on any component of the electrical system. The solution tank shall have a minimum capacity of 2 and 4 gallons for size 1 and 2 cleaners respectively, or the amount of cleaning solution required for 4 hours of operation. The cleaner shall be provided with a means for metering and injecting the solution into the water system. The means used for injecting the solution shall not result in progressive dilution of cleaning solution as the solution tank is emptied.

3.22.2 Dissolving of solution. The solution tank shall be equipped with connections and valves to permit filling the tank with water and for agitation of cleaning compound conforming to P-C-437. The means for agitation shall be the manufacturer's standard design.

3.22.3 Drainage. Means shall be provided for completely draining the solution tank. The drain connection shall include such piping and fittings as are required to provide ready accessibility for manual control of the discharge flow. The drain outlet shall be so located that the solution will not drain onto other components or structural members of the cleaner. The drain connection and any associated piping and fittings shall have a nominal pipe size of not less than 1/2 inch.

3.22.4 Measuring container. When specified (see 6.2), a measuring container with a capacity for not less than 60 ounces (oz) of cleaning compound shall be provided for each cleaner. The container shall be marked in increments of 20 oz to facilitate the addition of 20 oz of compound for every 2 gallons of water in the solution tank. Means shall be provided to secure the measuring container by clamps, hangers, or other devices in a location adjacent to the solution tank. The device for holding the container shall permit removal of the container without the use of tools. The capacity and markings of the measuring container shall be based on a cleaning compound weight of 60 pounds per cubic foot.

3.23 Compound metering device. The cleaner shall be equipped with a device to meter the compound introduced into the water system. The metering device shall permit adjustment of the compound flow rate from zero to not less than the metered capacity specified. When adjusted to the desired metering rate, the device shall automatically maintain a constant delivery rate of compound to within 10 percent of any tank levels under all conditions.

3.24 Delivery equipment. A vapor-cleaning gun 4-1/2 feet long, having a nozzle and shutoff valve, shall be furnished. Each gun assembly shall include an insulated swivel handle and air-cooled grips to protect operator from burns during operation. Nozzles shall be equipped with restrictors to supply flash steam at the nozzle outlet. No less than three nozzles shall be provided for use with the cleaning gun. The vapor nozzles shall include one round nozzle with a hardened tip, one 2-inch flat nozzle, and one 4-inch flat nozzle.

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3.24.1 Delivery hoses. A set of delivery hoses shall be provided for each steam cleaner. The hoses shall be suitable for vapor delivery. The set shall consist of one 50-foot and one 25-foot hose complete with all necessary couplings to connect both lengths of hose together and to make connections to the cleaner and the cleaner gun. The delivery hoses shall meet the construction and test requirements of MIL-H-28596 except that the male pipe-threaded hose nipples shall be held in place with bolting clamps and the size shall be 1/2-inch id.

3.24.2 Bypass or relief valve. A pressure bypass or relief valve shall be installed in the solution system to avoid the delivery hoses being subjected to excessive pressure. The bypass or relief valve shall have relieving capacity to keep the operating pressure of a cleaner within the safety limit.

3.24.3 Water supply hose. The water supply hose shall conform to grade I of ZZ-H-601, 3/4-inch id size, having an oil- and grease-resistant outer cover. The hose shall be 50 feet in length and shall be provided with male and female couplings having 3/4 inch by 11-1/2 inch (threads per inch) National Hose coupling threads conforming to ANSI B2.4.

3.25 Suction hose. When specified (see 6.2), 25 feet of suction hose having an id of not less than 1 inch shall be provided with each size 2 cleaner (see 3.10.6). The hose shall conform to grade V, type I of ZZ-H-561, and shall be provided with all necessary couplings conforming to WW-C-624. A foot check valve and strainer shall be provided.

3.26 Dimensions and weight. The dimensions and weight of the cleaner shall not exceed the following:

Height----------60 inches.
Width----------60 inches.
Length----------72 inches.
Total weight of each unit, including accessories: 1,700 pounds (lb).

3.27 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated as specified in section 5. The unit shall be conspicuously tagged to identify the lubricants and their temperature range. (All bearings (except sealed-for-life type), gears and sliding parts shall be provided with means for lubrication).

3.28 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other

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contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.29 Identification plate. An identification plate will be furnished by the contracting officer for each cleaner. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each cleaner in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.30 Instruction plates. The cleaner shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter. In addition, the cleaner shall have the following instructions:

a. "Use steam cleaning compound conforming to P-C-437 or in a concentration of 10 oz of compound per gallon of water in the solution tank." The instruction shall be marked in a conspicuous location on the exterior of the cleaner.
b. "Use light diesel oil, kerosene, or jet fuel." The instructions, to be in 1/2-inch lettering, shall be permanently and legibly marked on a noncorrosive plate to be attached adjacent to the fuel filler cap, and shall be marked on the outside front of the cleaner.

3.31 Maintenance tools. When specified (see 6.2), maintenance tools shall be furnished.

3.32 Workmanship. The quality of workmanship shall be such as to produce cleaners that are in accordance with the requirements of this specification and that are so constructed as to insure the proper functioning of all parts of the unit.

3.32.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.32.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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3.32.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.32.4 Welding. Welding procedures shall be in accordance with a nationally recognized ASME welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.32.5 Brazing and soldering. Brazing and soldering shall be in accordance with the manufacturer's standard practice. Resin-base fluxes shall be used in all soldering operations.

3.32.6 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.32.7 Forgings. Forgings shall be free from scale, enclosures, cold shuts, mismatching, and other defects that may affect the structural strength of the forgings. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste, shall not be attempted without authorization of the contracting officer.

3.32.8 Machine work. Tolerances for contact and bearing surfaces shall conform to standards prevailing among manufacturers normally producing steam cleaners.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. The first article inspection shall be performed on one cleaner when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.3, and the packaging inspection of 4.6.

4.3 Examination. Each cleaner shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Test conditions. The cleaner shall be tested and inspected under the following conditions:

4.4.1.1 Apparatus. When practicable, apparatus used in conjunction with testing specified herein shall be of the laboratory precision type and shall be calibrated at proper intervals to assure laboratory accuracy.

4.4.1.2 Fuel consumption. Fuel consumption measurement shall be made by the quantitative method, whereby the quantity consumed for a selected elapsed time shall be measured by a burette. The time selected shall be such that each reading will cover an elapsed time of at least 5 minutes. All expressions of quantities shall be recorded on a weight basis, and shall be measured by a direct weight-reading device such as a scale or precision meter calibrated to read in lb and oz, and shall be plotted on performance curves covering the entire operational range of the cleaner. Performance graphs shall include fuel consumption and thermal efficiency versus water flow rates.

4.4.1.3 Liquid flow rates. Liquid flow rates shall be determined by weighing the liquid flowing over a period of not less than 1 minute, or by measuring the time required to fill or empty a container of known volume. Flow-meter readings shall not be used, unless the flow-meter and use thereof has been specifically approved by the procuring activity.

4.4.1.4 Temperature. Temperatures shall be measured by means of appropriately located thermocouples or thermometers. When practicable, thermocouples shall be insulated from contact with other metals. Temperatures shall be expressed in degrees Fahrenheit.

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4.4.1.5 Pressure. Pressures shall be measured by properly calibrated Bourdon tube gages. The gages shall be of laboratory type having a dial diameter of not less than 4 inches.

4.4.2 Preparation for test. The cleaner shall be prepared for test by servicing with the fuels and lubricants specified by the manufacturer.

4.4.2.1 Running-in. The nature and extent of running-in shall be as specified by the manufacturer.

4.4.2.2 Discharge pressures. Discharge pressures specified herein shall be those measured at the discharge side of the heating coil.

4.4.2.3 Servicing cleaner. The cleaner shall be serviced prior to any testing, in accordance with the instructions of the manufacturer.

4.4.2.4 Cleaning compound. Cleaning compound conforming to P-C-437 or MIL-C-22542 shall be used in any and all tests specified herein.

4.4.2.5 Water supply. Ample water supply shall be connected to the water supply inlet on the cleaner unit.

4.4.2.6 Checking controls. All valves, the pump, and controls shall be checked to assure proper operation of the cleaner.

4.4.2.7 Data readings. During the testing specified herein, at least the following data shall be recorded at intervals not more than 30 minutes apart:

a. Date.
b. Time.
c. Ambient temperature.
d. Liquid flow rate (water or solution) in lb per minute.
e. Liquid temperature (water or solution) at outlet from heater.
f. Liquid temperature (water or solution) entering nozzle.
g. Liquid temperature (feedwater to cleaner) entering unit.
h. Water pressure at heater inlet.
i. Water pressure at heater outlet.
j. Pressure through heater.
k. Pressure drop to nozzle.
l. Heater fuel consumption in lb per hour.
m. Heater output in British thermal units per hour (Btu/hr).
n. Heater thermal efficiency.
o. API gravity of the fuel used.
p. Btu/lb of fuel.

Notes shall be entered in the logs regarding adjustments, servicing, vibrations, and any other irregularities and corrective measures taken during the test run.

4.4.2.8 Inspection, servicing, and adjustments. Planned stops shall be made at 8-hour intervals during the endurance test for inspections, servicing, and adjustments. Records shall be kept of any adjustments of a minor nature. Major adjustments shall be cause for rejection.

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4.4.2.9 Test time. Intervals of endurance test time of less than 1 hour, terminated by a parts failure, shall not be credited in 1/2-hour intervals, except where shorter periods are a test requirement.

4.4.3 Performance tests. The cleaner shall be subjected to the following tests:

4.4.3.1 Vapor spray test. The cleaner shall be operated for 6 hours as a vapor cleaner, and shall deliver the minimum cleaning solution per hour at design pressure (see 3.10.4).

4.4.3.2 Operating pressure test. The cleaner shall be operated at the specified rates and with feed water supplied within the normal temperature range of +40°F to +55°F to determine compliance with respect to:

a. Thermal efficiency (see 3.10.2).
b. Pressure regulation (see 3.10.4).

4.4.3.3 Impact performance test. The cleaner shall be checked for impact performance to determine compliance with 3.10.4. A 25-foot hose shall be used and the following method shall be applied: Impact (delivery force) values shall be derived by directing the vapor nozzle from a fixed position and distance as shown in figure 1, at a round target with 2 square inches of surface (1.6-inch id). The target device shall register the lb impact on a scale or gage without undue mechanical or actuating losses. The force, in lb, registered on the 2 square inch target shall be multiplied by 0.50 to determine the impact pressure in psi.

4.4.3.4 Safety controls test. Safety controls shall be tested to determine adequacy for protection of personnel operating the cleaner. The following tests shall be conducted:

4.4.3.4.1 Excess pressure test. While using the 50-foot hose, stop the solution at the gun. The pressure switch shall cut off the fuel and water before 150 lb of pressure is generated; hose connections shall not leak.

4.4.3.4.2 Water shortage or failure test. With the cleaner in full operation, turn off the water supply; the temperature control shall cut off the fuel to the burner when the temperature reaches 350°F.

4.4.3.5. Mobility test. The completely assembled cleaner with accessories shall be tested in accordance with 3.9.

4.4.3.6 Static pressure test. Subject the pressure system of each assembled cleaner to a hydrostatic test of not less than two times the maximum working pressure (see 3.10.4). Subject the burner coil to a hydrostatic test of not less than 1,000 psi. The pressure system and the burner coil shall show no evidence of leakage, damage, or deformation.

4.4.3.7 Dielectric strength test. The cleaner primary circuits shall be subjected to 1,000V RMS at 60 Hz, between mutually insulated circuits, and

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between insulated circuits and ground. Voltage shall be applied at the rate of approximately 500V per second until the specified voltage is reached. Voltage shall be maintained at 1,000V for a period of 1 minute to verify compliance with the requirement in 3.11.

4.5 Endurance test. When a first article is required, and unless otherwise specified (see 6.2), the first article shall be subjected to a 250-hour endurance run. The cleaner shall be operated with cleaning compound and run at the rated discharge pressure. The maximum efficiency loss shall not exceed 2 percent. The total 250 hours shall be accumulated in 8-hour runs in accordance with the manufacturer's operating procedures.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-S-12651. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-S-12651 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

* 6.1 Intended use. The cleaner covered by this specification is intended for the washing and cleaning of vehicles and miscellaneous ground equipment by the application of steam, pressure, and cleaning compound solution.

6.2 Ordering data. Acquisiton documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
d. Metal cabinet as specified shall be furnished (see 3.7).
e. Operating pressure, if other than as specified (see 3.10.4).
f. When a complete water system for size 2 cleaner shall be furnished as specified (see 3.10.6).
g. Requirement for electrical equipment, if other than as specified (see 3.11).
h. When measuring container shall be furnished as specified (see 3.22.4).
i. When suction hose is required (see 3.25).
j. When lubrication is required (see 3.27).
k. Color of finish coat shall be as specified (see 3.28).
l. When maintenance tools are required (see 3.31).
m. When cleaners are not to be tested in accordance with 4.5 (see 4.5).
n. Length of endurance test (hours) if other than as specified (see 4.5).
o. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.6).
p. Level of preservation and level of packing required (see 5.1).

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* 6.3 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one cleaner. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

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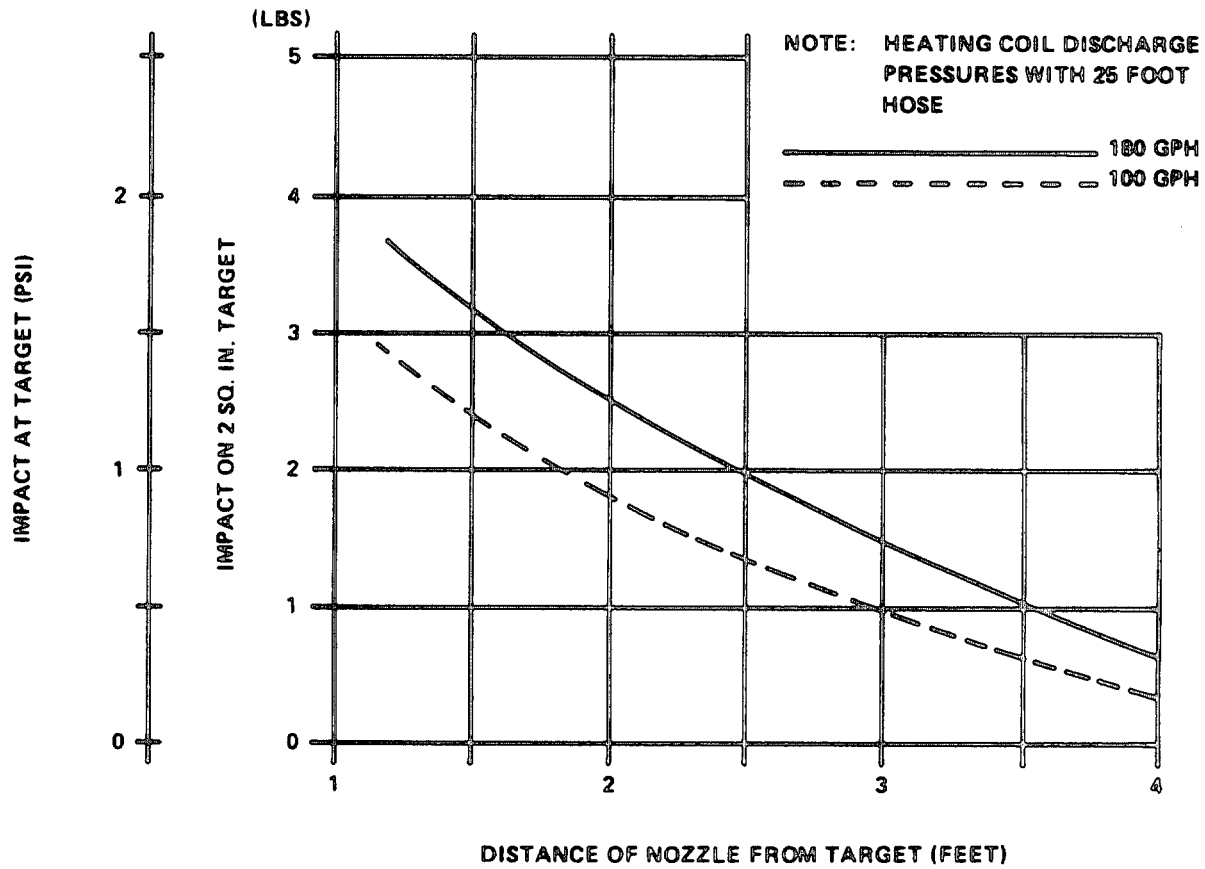
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(Project 4940-0475)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

4940-0475

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-22894E	CLEANER, STEAM, PRESSURE JET, WHEEL-MOUNTED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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